

Sequences and Series of Real Numbers (2005–2014)

Category: Real Analysis • Official Mock Test Series

TOTAL QUESTIONS

22

TOTAL MARKS

111 M

TEST DURATION

66 Min

PAPER PATTERN

Classic Paper Pattern (2

Section / Type		Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)		22	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2014 • Q19 • ID: JAM_2014_Q19)

MCQ

+2 M

Neg: -0.67

Q.19 Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two series, where $a_n = \frac{(-1)^n n}{2^n}$, $b_n = \frac{(-1)^n}{\log(n+1)}$ for all $n \in \mathbb{N}$. Then

- (A) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are absolutely convergent
- (B) $\sum_{n=1}^{\infty} a_n$ is absolutely convergent but $\sum_{n=1}^{\infty} b_n$ is conditionally convergent
- (C) $\sum_{n=1}^{\infty} a_n$ is conditionally convergent but $\sum_{n=1}^{\infty} b_n$ is absolutely convergent
- (D) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are conditionally convergent

Question 2 (JAM 2014 • Q18 • ID: JAM_2014_Q18)

MCQ

+2 M

Neg: -0.67

Q.18 Let $\{x_n\}$ be a sequence of real numbers such that $\lim_{n \rightarrow \infty} (x_{n+1} - x_n) = c$, where c is a positive real number. Then the sequence $\left\{\frac{x_n}{n}\right\}$

- (A) is NOT bounded
- (B) is bounded but NOT convergent
- (C) converges to c
- (D) converges to 0

Question 3 (JAM 2014 • Q17 • ID: JAM_2014_Q17)

MCQ

+2 M

Neg: -0.67

Q.17 Let $x_n = 2^{2n} \left(1 - \cos\left(\frac{1}{2^n}\right)\right)$ for all $n \in \mathbb{N}$. Then the sequence $\{x_n\}$

- (A) does NOT converge
- (B) converges to 0
- (C) converges to $\frac{1}{2}$
- (D) converges to $\frac{1}{4}$

Question 4 (JAM 2014 • Q6 • ID: JAM_2014_Q6)

MCQ

+1 M

Neg: -0.33

Q.6 The radius of convergence of the power series $\sum_{n=0}^{\infty} 2^{2n} x^{n^2}$ is

- (A) $\frac{1}{4}$ (B) 1 (C) 2 (D) 4

Question 5 (JAM 2013 • Q7 • ID: JAM_2013_Q7)

MCQ

+2 M

Neg: -0.67

Let

$$x_n = \left(1 - \frac{1}{3}\right)^2 \left(1 - \frac{1}{6}\right)^2 \left(1 - \frac{1}{10}\right)^2 \cdots \left(1 - \frac{1}{\frac{n(n+1)}{2}}\right)^2, \quad n \geq 2.$$

Then $\lim_{n \rightarrow \infty} x_n$ is

- (A) $\frac{1}{3}$ (B) $\frac{1}{9}$ (C) $\frac{1}{81}$ (D) 0

Question 6 (JAM 2012 • Q13 • ID: JAM_2012_Q13)

MCQ

+6 M

Neg: -2.0

Q.13 If the power series $\sum_{n=0}^{\infty} a_n x^n$ converges for $x = 3$, then the series $\sum_{n=0}^{\infty} a_n x^n$

- (A) converges absolutely for $x = -2$
(B) converges but not absolutely for $x = -1$
(C) converges but not absolutely for $x = 1$
(D) diverges for $x = -2$

Question 7 (JAM 2012 • Q1 • ID: JAM_2012_Q1)

MCQ

+6 M

Neg: -2.0

Q.1 Let $\{x_n\}$ be the sequence $+\sqrt{1}, -\sqrt{1}, +\sqrt{2}, -\sqrt{2}, +\sqrt{3}, -\sqrt{3}, +\sqrt{4}, -\sqrt{4}, \dots$.
If

$$y_n = \frac{x_1 + x_2 + \dots + x_n}{n} \text{ for all } n \in \mathbb{N},$$

then the sequence $\{y_n\}$ is

- (A) monotonic
- (B) NOT bounded
- (C) bounded but NOT convergent
- (D) convergent

Question 8 (JAM 2011 • Q14 • ID: JAM_2011_Q14)

MCQ

+6 M

Neg: -2.0

Q.14 The set of all x at which the power series $\sum_{n=1}^{\infty} \frac{n}{(2n+1)^2} (x-2)^{3n}$ converges is

- (A) $[-1, 1)$
- (B) $[-1, 1]$
- (C) $[1, 3)$
- (D) $[1, 3]$

Question 9 (JAM 2011 • Q3 • ID: JAM_2011_Q3)

MCQ

+6 M

Neg: -2.0

Q.3 The value of $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sqrt{n^2 + kn}}$ is

- (A) $2(\sqrt{2}-1)$
- (B) $2\sqrt{2}-1$
- (C) $2-\sqrt{2}$
- (D) $\frac{1}{2}(\sqrt{2}-1)$

Question 10 (JAM 2011 • Q1 • ID: JAM_2011_Q1)

MCQ

+6 M

Neg: -2.0

Q.1 Let $a_n = \sum_{k=1}^n \frac{n}{n^2 + k}$, for $n \in \mathbb{N}$. Then the sequence $\{a_n\}$ is

- (A) Convergent
- (B) Bounded but not convergent
- (C) Diverges to ∞
- (D) Neither bounded nor diverges to ∞

Question 11 (JAM 2010 • Q13 • ID: JAM_2010_Q13)

MCQ

+6 M

Neg: -2.0

Q.13 The radius of convergence of the power series $\sum_{n=0}^{\infty} a_n z^{n^2}$, where $a_0 = 1$, $a_n = 3^{-n} a_{n-1}$ for $n \in \mathbf{N}$, is

- (A) 0 (B) $\sqrt{3}$ (C) 3 (D) ∞

Question 12 (JAM 2010 • Q1 • ID: JAM_2010_Q1)

MCQ

+6 M

Neg: -2.0

Q.1 Which of the following conditions does NOT ensure the convergence of a real sequence $\{a_n\}$?

- (A) $|a_n - a_{n+1}| \rightarrow 0$ as $n \rightarrow \infty$
 (B) $\sum_{n=1}^{\infty} |a_n - a_{n+1}|$ is convergent
 (C) $\sum_{n=1}^{\infty} n a_n$ is convergent
 (D) The sequences $\{a_{2n}\}$, $\{a_{2n+1}\}$ and $\{a_{3n}\}$ are convergent

Question 13 (JAM 2009 • Q9 • ID: JAM_2009_Q9)

MCQ

+6 M

Neg: -2.0

Q.9 The set of all positive values of a for which the series $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \tan^{-1} \left(\frac{1}{n} \right) \right)^a$ converges, is

- (A) $\left(0, \frac{1}{3}\right]$ (B) $\left(0, \frac{1}{3}\right)$ (C) $\left[\frac{1}{3}, \infty\right)$ (D) $\left(\frac{1}{3}, \infty\right)$

Question 14 (JAM 2009 • Q7 • ID: JAM_2009_Q7)

MCQ

+6 M

Neg: -2.0

Q.7 The set of all limit points of the sequence $1, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}, \frac{1}{16}, \frac{3}{16}, \frac{5}{16}, \frac{7}{16}, \frac{9}{16}, \dots$ is

- (A) $[0, 1]$
 (B) $(0, 1]$
 (C) the set of all rational numbers in $[0, 1]$
 (D) the set of all rational numbers in $[0, 1]$ of the form $\frac{m}{2^n}$ where m and n are integers

Question 15 (JAM 2009 • Q6 • ID: JAM_2009_Q6)

MCQ

+6 M

Neg: -2.0

Q.6 Let $a_n = \begin{cases} \frac{1}{3^n} & \text{if } n \text{ is a prime,} \\ \frac{1}{4^n} & \text{if } n \text{ is not a prime.} \end{cases}$

Then the radius of convergence of the power series $\sum_{n=1}^{\infty} a_n x^n$ is

- (A) 4 (B) 3 (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

Question 16 (JAM 2008 • Q12 • ID: JAM_2008_Q12)

MCQ

+6 M

Neg: -2.0

Q.12 Let $\{a_n\}$ and $\{b_n\}$ be sequences of real numbers defined as $a_1=1$ and for $n \geq 1$,
 $a_{n+1} = a_n + (-1)^n 2^{-n}$, $b_n = \frac{2a_{n+1} - a_n}{a_n}$. Then

- (A) $\{a_n\}$ converges to zero and $\{b_n\}$ is a Cauchy sequence
(B) $\{a_n\}$ converges to a non-zero number and $\{b_n\}$ is a Cauchy sequence
(C) $\{a_n\}$ converges to zero and $\{b_n\}$ is not a convergent sequence
(D) $\{a_n\}$ converges to a non-zero number and $\{b_n\}$ is not a convergent sequence

Question 17 (JAM 2008 • Q3 • ID: JAM_2008_Q3)

MCQ

+6 M

Neg: -2.0

Q.3 Let $s_n = \int_0^1 \frac{n x^{n-1}}{(1+x)} dx$ for $n \geq 1$. Then as $n \rightarrow \infty$, the sequence $\{s_n\}$ tends to

- (A) 0 (B) $1/2$ (C) 1 (D) $+\infty$

Question 18 (JAM 2007 • Q10 • ID: JAM_2007_Q10)

MCQ

+6 M

Neg: -2.0

10. Suppose (c_n) is a sequence of real numbers such that $\lim_{n \rightarrow \infty} |c_n|^{1/n}$ exists and is non-zero.

If the radius of convergence of the power series $\sum_{n=0}^{\infty} c_n x^n$ is equal to r , then the radius of convergence of the power series $\sum_{n=1}^{\infty} n^2 c_n x^n$ is

- (A) less than r
- (B) greater than r
- (C) equal to r
- (D) equal to 0

Question 19 (JAM 2007 • Q7 • ID: JAM_2007_Q7)

MCQ

+6 M

Neg: -2.0

7. Let (a_n) be an increasing sequence of positive real numbers such that the series $\sum_{k=1}^{\infty} a_k$ is

divergent. Let $s_n = \sum_{k=1}^n a_k$ for $n = 1, 2, \dots$ and $t_n = \sum_{k=2}^n \frac{a_k}{s_{k-1}s_k}$ for $n = 2, 3, \dots$. Then $\lim_{n \rightarrow \infty} t_n$ is equal to

- (A) $1/a_1$
- (B) 0
- (C) $1/(a_1 + a_2)$
- (D) $a_1 + a_2$

Question 20 (JAM 2006 • Q1 • ID: JAM_2006_Q1)

MCQ

+6 M

Neg: -2.0

1. $\lim_{n \rightarrow \infty} \frac{2^{n+1} + 3^{n+1}}{2^n + 3^n}$ equals

- (A) 3
- (B) 2
- (C) 1
- (D) 0

Question 21 (JAM 2005 • Q14 • ID: JAM_2005_Q14)

MCQ

+6 M

Neg: -2.0

14. Let $f: (-1, 1) \rightarrow \mathbf{R}$ be such that $f^{(n)}(x)$ exists and $|f^{(n)}(x)| \leq 1$ for every $n \geq 1$ and for every $x \in (-1, 1)$. Then f has a convergent power series expansion in a neighbourhood of

- (A) every $x \in (-1, 1)$
- (B) every $x \in \left(-\frac{1}{2}, 0\right)$ only
- (C) no $x \in (-1, 1)$
- (D) every $x \in \left(0, \frac{1}{2}\right)$ only

Question 22 (JAM 2005 • Q1 • ID: JAM_2005_Q1)

MCQ

+6 M

Neg: -2.0

1. Let $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$ be sequences of real numbers such that $b_n = a_{2n}$ and $c_n = a_{2n+1}$. Then $\{a_n\}$ is convergent

- (A) implies $\{b_n\}$ is convergent but $\{c_n\}$ need not be convergent
- (B) implies $\{c_n\}$ is convergent but $\{b_n\}$ need not be convergent
- (C) implies both $\{b_n\}$ and $\{c_n\}$ are convergent
- (D) if both $\{b_n\}$ and $\{c_n\}$ are convergent

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Sequences and Series of Real Numbers (2005–2014) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+2	B
2	MCQ	+2	C
3	MCQ	+2	C
4	MCQ	+1	B
5	MCQ	+2	B
6	MCQ	+6	A
7	MCQ	+6	D
8	MCQ	+6	C
9	MCQ	+6	A
10	MCQ	+6	A
11	MCQ	+6	B

Q. No.	Type	Marks	Official Key
12	MCQ	+6	A
13	MCQ	+6	C
14	MCQ	+6	A
15	MCQ	+6	B
16	MCQ	+6	B
17	MCQ	+6	B
18	MCQ	+6	C
19	MCQ	+6	A
20	MCQ	+6	A
21	MCQ	+6	A
22	MCQ	+6	C

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.